UNIT 19 CONTENT ANALYSIS

Structure

19.0 OBJECTIVES

After going through this Unit, you will be able to:

- explain the concept and meaning of the content analysis;
- identify the areas of research in social sciences wherein content analysis can be used;
- discuss the various types of content analysis;
- state the various approaches of content analysis;
- describe the various steps involved in using the content analysis;
- enable you to analyse and interpret the results; and
- discuss the advantages and disadvantages of the content analysis.

19.1 INTRODUCTION

Many research Scholars take up historical overviews of economics as an area of research. They probe the research questions like how the different streams of economics like institutional economics, Neo-classical economics, behavioural economics etc. have evolved over a period of time and how researcher's focus had shifted from one area to another isolating a particular paradigm change. Sometimes a researcher aims to lay concepts and meanings from the social actors' account of realities related to his identified research problem. They use the sources like recorded interviews, letters, journals, newspaper stories and verbal materials. In such situations, content analysis as data (information) analysis technique is used by the social scientists. Much of the subject matter of social sciences including consumer studies is in the form of verbal and nonverbal behaviour. The exchange process in the market place and the communication of the values of the exchange depends upon the written or spoken words. Much of consumer research has concentrated on the characteristics, opinion, or behaviour of the interpreter of communication messages. In media economics, issues like monopolistic competition in TV channels, competition and diversity in newspaper industry, effects of group ownership on daily newspaper content etc. can be probed by content analysis alongwith other statistical techniques. Similarly content analysis can also be used to analyse the cultural issues in cultural economics to capture pattern form of explanation. Hence, in this unit, we will discuss the concept of content analysis, its various types and approaches, the procedures involved in its application, and its advantages and disadvantages. Let us begin with historical background of content analysis followed by describing its concept and meaning.

19.2 HISTORICAL BACKGROUND OF CONTENT ANALYSIS

We find different approaches to analysis and comparison of texts in hermeneutic contexts (e.g. bible interpretations) like newspaper analysis, graphological procedures, content analysis, and the dream analysis by Freud etc. The basis of quantitative content analysis had been laid by Lazarsfield and Lasswell in the USA during 1920's and 30's. Historically, content analysis was a time consuming process. Analysis was done either manually, or by slow mainframe computers to analyze punch cards containing data punched by human coders. Single study could employ thousands of these cards. Human error and time constraints made this method impractical for large texts. However, despite its impracticality, content analysis was utilized as research method by 1940's. Although initially limited to studies that examined texts for the frequency of the occurrence of identified terms (word counts), by mid-1950's researchers started to consider the need for more sophisticated methods of analysis, focusing on concepts rather than simply words, and on semantic relationships rather than just presence. The first textbook about this method was published by Berelson in 1952. In the sixties of 20th century, content analysis found its way into linguistics, psychology, sociology, history, arts etc. The procedures involved in content analysis were refined by incorporating models of communication; analysis of non-verbal aspects, contingency analysis, computer applications. Since the middle of 20th century objections had been raised against a superficial analysis without respecting latent contents and contexts, working with simplifying and distorting quantification (Kracauer, 1952). Subsequently qualitative approaches to content analysis had been developed. Today, the use of modern computers makes content analysis much simpler and the data less vulnerable to human error.

19.3 CONTENT ANALYSIS: CONCEPT AND MEANING

Content analysis is a set of procedures for collecting and organizing information in a standardized format that allows analysts to make inferences about the characteristics and meaning of written and other recorded materials.

Content analysis is a multipurpose research method developed specifically for investigating a broad spectrum of problems in which the content of communication serves as the basis of inference. Content analysis covers both the content of the material and its structure. Content refers to the specific topics or themes in the material. Structure refers to form. Whether an article is prominently featured on the first page of a newspaper or buried in the middle section is a structural question. Careful reading of the written materials is necessary for all kinds of researches in social sciences. But content analysis is something different from careful reading of written materials in many ways.

According to **Berelson**, content analysis is a research technique for the objective, systematic, and quantitative description of the manifest content of communication. The term content analysis is used here to mean the scientific analysis of communication messages. The method being "scientific" catholic in nature, requires that the analysis be rigorous and systematic. According to **Paisley**, content analysis is a phase of information processing in which communication content is transformed through objective and systematic application of categorization rules into data that can be summarized and compared.

This definition underlines the point that content analysis is a systematic technique for analyzing message content and message handling. It is a tool for observing and analyzing the overt communication behaviour of selected communicators.

Content analysis , while certainly a method of analysis, is more than that, it is a method of observation. Instead of observing people's behaviour directly, or asking them to respond to scales or interviewing them, the investigator takes the communications that people have produced and asks questions of the communication (Kerlinger, 1968, p.544).

Content analysis refers to any procedure for answering the relative extent to which specified references, attitudes or themes permeate a given message or document (Stone, 1964).

Neuendorf defines content analysis as " the systematic, objective, quantitative analysis of message characteristics".

Based on the above definitions, the following characteristics of content analysis emerge: objectivity, systematic and generality.

Objectivity: To have objectivity, the analysis must be carried out on the basis of explicitly formulated rules which will enable two or more investigators to obtain the same results from the same documents. This requirement of objectivity gives scientific standing to content analysis and differentiates it from literary crtiticism.

Systematic: In a systematic analysis, the inclusion and exclusion of content or categories is done according to consistently applied criteria of selection. This requirement is meant to eliminate elements in the content which do not fit in the analyst's thesis.

Generality: By generality, we mean that the findings must have theoretical relevance, purely descriptive information about content, unrelated to other attributes of content or to the characteristics of communicator or recipient of the message, is of little scientific value.

Thus, content analysis can be defined as a research technique for collecting and organizing information in a standardized format for making inference systematically and objectively about the characteristics of message.

19.4 TERMS USED IN CONTENT ANALYSIS

The following concepts and terms are frequently used in content analysis. Let us understand these terms.

Manifest Content Analysis: It involves simply counting words, phrases, or "surface" features of the text itself. It provides reliable quantitative data that can easily be analyzed using inferential statistics.

Latent Content Analysis: It involves interpreting the underlying meaning of the text. Latent analysis is different to manifest analysis because researcher must have a clearly stated idea about what is being measured. The value of the latent content analysis actually depends upon the researcher's ability to expose previously marked themes, messages and cultural values within the text. This analysis is widely used in qualitative content analysis.

Content Units: There are two types of content units i.e. the unit of analysis and unit of observation. The unit of analysis concerns the general idea or phenomenon being studied. The unit of observation concerns the specific item measured at an individual level.

Coding: Coding is the process whereby raw data are systematically transformed and aggregated into units which permit precise description of relevant content characteristics. There are two methods of coding: (i) deductive measurement, (ii) inductive measurement.

Deductive Measurement: It requires the development of specific coding categories before a researcher starts a content analysis. Deductive measurement is useful with an established set of coding categories or if a clear hypothesis or research question exists at the outset of the analysis.

Inductive Measurement: This method supports the practice of emergent coding, which means that the basic research question or hypothesis for a formal content analysis emerges from the units of observation. It entails creating coding categories during the analysis process. Emergent coding is useful in exploratory content analysis.

Coding Scheme: Every analysis begins with an existing coding scheme. Coding scheme is another phrase of coding categories and coding book, within which all instances of the content are analyzed or noted. Every coding scheme consist of the following statements.

Master Code Book: It provides the coders with explicit instructions and defines each word/ phrase/ aspect to be analyzed. It explains how to use code sheet.

Code Sheet: Code sheet provides the coders with a form on which they note every instance of every word/ phrase/ aspect being analyzed. The code sheet lists all the coding categories.

Coding Dictionary: It is used with computer based content analysis. A set of words/ phrases, part of speech …. That is used as the basis for a search text.

Check Your Progress 1

1) Identify the areas of research in economics wherein content analysis can be used.

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2) What is the distinction between manifest content analysis and latent content analysis?

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19.5 APPROACHES OF CONTENT ANALYSIS

Instead of being a single technique, content analysis is a collection of different approaches to the analysis of texts or more generally of messages of any kind. Important approaches frequently used in Social Sciences are discussed below:

19.5.1 Conceptual Content Analysis

Traditionally, content analysis has most often been thought in terms of conceptual analysis. In conceptual analysis, a concept is chosen for examination and the analysis involves quantifying and tallying its presence. It is also known as thematic analysis. The focus here is on looking at the occurrence of selected terms within a text or texts, although the terms may be implicit as well as explicit. While explicit terms obviously are easy to identify, coding for implicit terms and deciding their level of implication is complicated by the need to base judgments on a somewhat subjective system.

19.5.2 Relational Content Analysis

Relational content analysis, like conceptual analysis, begins with the act of identifying concepts present in a given text or set of texts. However, relational analysis seeks to go beyond presence by exploring the relationships between the concepts identified. Relational analysis has also been termed semantic analysis.

In other words, the focus of relational analysis is to look for semantic, or meaningful relationships. Individual concepts, in and of themselves, are viewed as having no inherent meaning. Carley (1992) asserts that concepts are 'ideational kernels.' These kernels can be thought of as symbols which acquire meaning through their connections to other symbol. Relational content analysis approach can be of three types:

i) **Affect extraction:** This approach provides an emotional evaluation of concepts explicit in a text. It is problematic because emotion may vary across time and populations. Nevertheless, when extended, it can be a potent means of exploring the emotional/psychological state of the speaker and/or writer.

ii) **Proximity analysis:** This approach, on the other hand, is concerned with the co-occurrence of explicit concepts in the text. In this procedure, the text is defined as a string of words. A given length of words, called a *window*, is determined. The window is then scanned across a text to check for the co-occurrence of concepts. The result is the creation of a concept determined by the *concept matrix*. In other words, a matrix, or a group of interrelated, co-occurring concepts, might suggest a certain overall meaning.

iii) **Cognitive mapping:** This approach is one that allows for further analysis of the results from the two previous approaches. It attempts to take the above processes one step further by representing these relationships visually for comparison. Whereas affective and proximal analysis function primarily within the preserved order of the text, cognitive mapping attempts to create a model of the overall meaning of the text. This can be represented as a graphic map that represents the relationship between concepts. In this manner, cognitive mapping lends itself to the comparison of semantic connections across texts. This is known as map analysis which allows for comparisons to explore "how meanings and definitions shift across people and time".

19.6 PROCEDURE INVOLVED IN CONTENT ANALYSIS

The various steps involved in content analysis are: (i) Formulation of research questions, (ii) Determining materials to be included, (iii) Developing content categories, (iv) Selecting and Finalizing units of analysis, (v) Code the materials, (vi) Analyze and interpret the results. These are discussed below:

19.6.1 Formulation of Research Questions

Content analysis begins with a specific statement of the objectives or research questions. The objective of content analysis is to convert recorded "raw" phenomenon into data, which can be treated essentially in a scientific manner so that body of knowledge may be built up. Objectives are precisely worded questions that investigator/s is/are trying to answer. By making a clear statement of the research question, the researcher can ensure that the analysis focuses on those aspects of content which are relevant for the research. The question should be based on a clear understanding of research needs and the available data. Therefore, the selection of the topic should be such that can be answered by analyzing the appropriate communication content.

In general content analysis can be used to answer " What" but not " Why" question. It helps analysts to describe or summarize the content of written material , the attitudes or perceptions of its writers, or its effects on the audience. For example, if analysts want to assess the effects of different women empowerment programmes on the lives of younger and middle aged women in rural and urban area. Content analysis of open–ended interview / responses could be used to identify their outlook, attitudes and security about their life.

19.6.2 Determining Materials to be Included

The next step of content analysis is to decide relevant communication content to answer the research question and to determine the time period to be covered. Content analysis can be used to study any recorded materials as long as the information is available to be reanalyzed for reliability checks. It is used most frequently to analyze written materials to study any recorded communication including TV programmes, movies, photographs, regulations, other public documents, workplaces, case studies, reports, answer to survey questions, newspapers, news release, books, Journals article, letters etc. Speeches and discussions can also be analyzed.

Sampling: Sampling is necessary if the population is too extensive to be analyzed. Thus a sample should be selected from the population in order to make valid conclusion and generalization about a population. Simple random sampling, interval sampling, cluster sampling and multistage sampling techniques are used in content analysis.

19.6.3 Developing Content Categories

Content analysis is no better than its categories., since they reflect the formulated thinking , hypotheses, and the purpose of the study. Categories provide structure for grouping and recording units. Developing the category system to classify the text is the heart of content analysis. Berelson (1952) has emphasized the importance of formulating coding categories by quoting that " content analysis stands or falls by its categories. Particular studies have been productive to the extent that categories were clearly formulated and well adequate to the problem and to the content".

To be useful, every content category must be thoroughly defined, indicating what type of material be and not to be included. Chadwick et.al. (1984) have also emphasized the following three characteristics of content categories i.e. (i) Categories should be exhaustive so that all relevant items in the material being studied can be placed within a category. (ii) It should be mutually exclusive so that no item can be coded in more than one category. (iii) Thirdly categories should be independent so that recording of unit's category assignment is not affected by the category assignment of other recording units.

Category format: Categories can be conceptualized in many ways. Some common category formats are grouping, scales, and matrices. Structured category format increase coding efficiency especially when the number of categories are large. Scales provide the rank ordering information. Matrices are useful formats when analysts seek more information about issues than simply whether they are present or absent.

19.6.4 Selecting Units of Analysis

Once the categories have been identified, the analyst would be interested in determining the unit of content for classification under the content categories and the system of enumeration for the same. The unit of analysis is the smallest unit of content that is coded under the content category. The unit of analysis vary with the nature and objective of the analysis. Thus, the unit of analysis might be a single word, a theme, a letter, a symbol, a news story, a short story, a character or an entire article etc. There are two kinds of unit of analysis : *Recording unit and Context unit.*

The Recording Unit is the specific segment of content in which the occurance of a referene /fact is counted or the unit can be broken down so that reference/ facts can be placed in different categories.

The Context Unit is the larger body of the content that may be searched to characterize the recording unit. Context units set limits on the portion of written material that is to be examined for categories of words or statements. A recording unit is the specific segment of context unit in written material that is placed in a category. For example, if the coding unit is the word, then the context unit may be the sentence or the pargraph in which the word appears and characterizes the recording unit.

Word: The smallest unit generally used in the content analysis as a unit is a word. Lasswell(1952) calls word as a symbol and may include word compounds e,g, phrases as well as single word. In this type of research one might study the relative occurrence of key symbols or value laden terms until the content has been systematically examined relevant to the hypotheses of the study.

Theme: The theme is a single assertion about a subject. The theme is among the most useful units of content analysis because issues, values, beliefs, and attitudes are usually discussed in this form.

Character: Character may be defined as a use of fictional or historical character as the recording unit is also employed.

Item: The item is the whole natural unit employed by procedures of symbolic material. It may be the entire speech, radio programme, letter to the editors, or news story.

Space and Time Measures: Some studies have classified content by physical division such as column inch, the line or paragraph, the minutes or the foot film.

Finalizing units of analysis: In content analysis, the counting or quantification of the units is performed by using three methods of enumeration : 1) Frequency, 2) Intensity or Direction, and Space/ Time.

Frequency: Frequency simply means counting whether or not something occurs and how often (how many times).

Direction: Direction is nothing but the direction of messages in the content along some continuam e,g, positive, negative, supporting or opposed.

Intensity: Intensity is the strength or power of a message in a direction. For example, the characteristic of forgetfulness can be minor (e.g. not remembering to take the keys when leaving home, taking time to recall the name of someone whom you have not seen in years) or major (e.g. not remembering your name, not recognizing your children).

Space: A researcher can record the size of the text messages or the amount of space or volume allocated to it. Space in written text is measured by counting words, sentences, paragraphs, or space on a page (e.g. square inches) for video or audio text. Space can also be measured by the amount of time allotted.

To explain the differences among the above described quantification levels and how they relate to constructing categories, let us give a hypothetical example of 'Importance of FDI in public sector organizations'. The analyst has a major source of information as newspaper, articles, public documents, transcripts of interview with political leaders, and public officials. For each issue of each newspaper in the sample, the analysts add together number of column inches from all news articles/editorials to find he total number of space for each position in addition to the coding, the name , location, and date of each newspaper. The analyst who use this level of quantification have to assure that the differences they find in amounts of space are valid indicators of relative emphasis or importance. At the next level of quantification, the analyst can code the frequency of recording units by tallying the number of times each issues or statements occurs in the text. At the third level of quantification, analyst provide code for intensity. Frequencies are counted but each coded statement or issues is also adjusted by a weight that measures related intensity.

19.6.5 Code the Materials

Coding the unit of analysis into content category is called coding. Defining categories and preparing coding schedule for analysis and coding of the content are almost simultaneous steps. Material can be coded either manually or by computers, depending on the sources available and format of the material. After deciding how the material will be coded, the analyst writes the instructions for coding. According to Kripendorff (1980), the guidelines for coding instructions include definition of recording units and procedures for identifying theme, descriptions of variables and categories, outline of cognitive procedures used in placing data in categories, and instructions for using and administering data sheets.

Pre-testing: Pretesting is an important step before actual coding begins. It involves coding a small portion of the material to be analyzed. A pretest enables the analyst to determine whether (1) the categories are clearly specified and meet the necessary requirement, (2) the coding instructions are adequate and, (3) the coders are suitable for coding. These are determined on the basis of reliability among coders band consistency in individual coding decision. If analyst find that material can be coded with high reliability then actual coding begins.

19.6.6 Analyzing and Interpreting the Results

The main objective of content analysis is to analyze the collected information with regard to the proposed objective of the analysis. The analysis involves summarizing the coded data, discovering patterns and relationships within the data, testing hypotheses about the patterns and relationship to assess the validity of the analysis.

The absolute and relative frequencies are most commonly used for summarizing the data. Absolute frequency might be the number of times statements or issues are found in the sample. A relative frequency might be represented by a percentage of the sample size.

Another way of analyzing content analysis is to examine relations among variables by cross tabulating the cooccurrence of variables. Other techniques for discovering patterns of relationship in data include contingency analysis, cluster analysis and factor analysis.

One important development in analyzing content analysis of the data is the use of computer. Computer programme like "General Inquirer" can identify within a body of text , those words and phrases that belong to specified categories (Stone et.al., 1996). Other computer programme like SPSS, Atlas titi, Minitab etc. are also very useful in both quantitative and qualitative content analysis.

Whatever the technique used, a final and important task is to assess the validity of the result by relating them to other data that are known to reasonably valid. The inference a researcher can or cannot make on the basis of research is critical in content analysis. Content analysis describes what is in the text. It cannot reveal the intentions of those who created the text or the effects that messages in the text have on those who receive them.

Check Your Progress 2

1) State the different sources of content analysis.

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2) Which methods are used to quantify the units of analysis?

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3) Briefly describes the different steps involved in the process of content analysis.

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